

ADVANCE TRAUMA

Statistical Analysis Plan

Effects of Advanced Trauma Life Support®
Training Compared to Standard Care on Adult
Trauma Patient Outcomes: A Stepped-Wedge
Cluster Randomised Trial

Version 0.7.0, 2026-08-04

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1 Administrative information

1.1 Study identifiers

- Protocol version 1.5.0 dated 2025-06-12
- ClinicalTrials.gov identifier: [NCT06321419](#)
- Clinical Trials Registry - India identifier: CTRI/2024/07/071336

1.2 Changelog

We will add a changelog here once we finalise version 1.0.0.

1.3 Contributors

Name and ORCID	Affiliation	Role
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1.4 Use of generative artificial intelligence

We used generative artificial intelligence tools, including large language models accessed via Cursor, to help draft, revise, and format this statistical analysis plan. We reviewed and approved all substantive statistical content, and we take full responsibility for the final document.

2 Introduction

2.1 Trial synopsis

Title Effects of Advanced Trauma Life Support® Training Compared to Standard Care on Adult Trauma Patient Outcomes: A Stepped-Wedge Cluster Randomised Trial

Rationale Trauma is a massive global health issue. Many training programmes have been developed to help physicians in the initial management of trauma patients. Among these programmes, Advanced Trauma Life Support® (ATLS®) is the most popular, having trained over one million physicians world-wide. Despite its widespread use, there are no controlled trials showing that ATLS® improves patient outcomes. Multiple systematic reviews emphasise the need for such trials.

Aim To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

Primary Outcome In-hospital mortality within 30 days of arrival at the emergency department.

Trial Design Batched stepped-wedge cluster randomised trial in India.

Trial Population Adult trauma patients presenting to the emergency department of a participating hospital.

Sample Size 30 clusters and at least 4320 patients.

Eligibility Criteria

Clusters are secondary or tertiary hospitals providing trauma care in India that admit or refer/transfer for admission at least 400 patients with trauma per year. Within each cluster, one or more units of physicians providing initial trauma care in the emergency department will be trained.

Participants are adult trauma patients who present to the emergency department of participating hospitals and are admitted or transferred for admission.

Intervention The intervention will be ATLS® training, a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. Physicians will be trained in an accredited ATLS® training facility in India.

Ethical Considerations We will use an opt-out consent approach for collection of routinely recorded data. We will obtain informed consent for collection of non-routinely recorded data, such as quality of life and disability outcomes. Patients who are unconscious or lack a legally authorized representative will be included under a waiver of informed consent. Note that consent here refers to consent to data collection.

Trial Period February 2025 to October 2029

2.2 Special considerations

2.3 Funding

This trial is not yet fully funded. The Trial Management Group has decided to proceed with the trial with the expectation that additional funding will be secured. The Trial Steering Committee will be informed of the funding status at each meeting. If funding is not secured, the trial will be stopped. This will likely result in an underpowered trial. The justification for this decision is that the intervention is considered standard of care in many countries and the data collection is considered minimal risk. There is therefore a very small risk of harm to patient participants, but a potential direct benefit to those patient participants who receive the intervention. The benefit-risk ratio is therefore considered to be favourable, even in the case of an underpowered trial.

3 Aim and objectives

3.1 Aim

To compare the effects of ATLS® training with standard care on outcomes in adult trauma patients.

3.2 Objectives

- To estimate the effects of ATLS® training on 30-day in-hospital mortality compared to standard care.
- To assess how robust the main analysis results are to different model specifications, exposure definitions, and missing-data assumptions, through sensitivity analyses.
- To estimate the effects of ATLS® training on secondary outcomes compared to standard care.
- To explore the effects of ATLS® training on subgroups of patients compared to standard care.

3.3 Framework

We have designed and will analyse the trial within a superiority framework, with ATLS® training compared with standard care for the primary and secondary outcomes. We target a cluster-specific (conditional) intervention effect among eligible patient participants, not a marginal (population-averaged) effect and not an equal-weight average across clusters.

4 Study design

4.1 Trial design

This is a batched stepped-wedge cluster randomised trial with 30 clusters (see Figure 1). Each cluster is one hospital, and we randomise hospitals, even when we train more than one unit of physicians in the same hospital. There are two intervention conditions: ATLS® training and standard care.

The trial comprises 6 batches of identical 13-period 5-sequence designs, with one cluster allocated to each sequence of each batch (equal allocation across sequences)¹. Each period is one month, and each cluster will be in the trial for a total of 13 months. We measure clusters repeatedly across periods. We will implement the intervention during a one-month transition period, which we exclude from the analysis. There will be an anticipated overlap of 6 months between successive batches.

Clusters will be sequentially allocated to batches based on when they enter the study, and within each batch clusters will then be randomised to an intervention implementation sequence. We include individual patients cross-sectionally for the primary outcome, although we assess some secondary outcomes at multiple follow-up time points. The anticipated trial period is from February 2025 to October 2029.

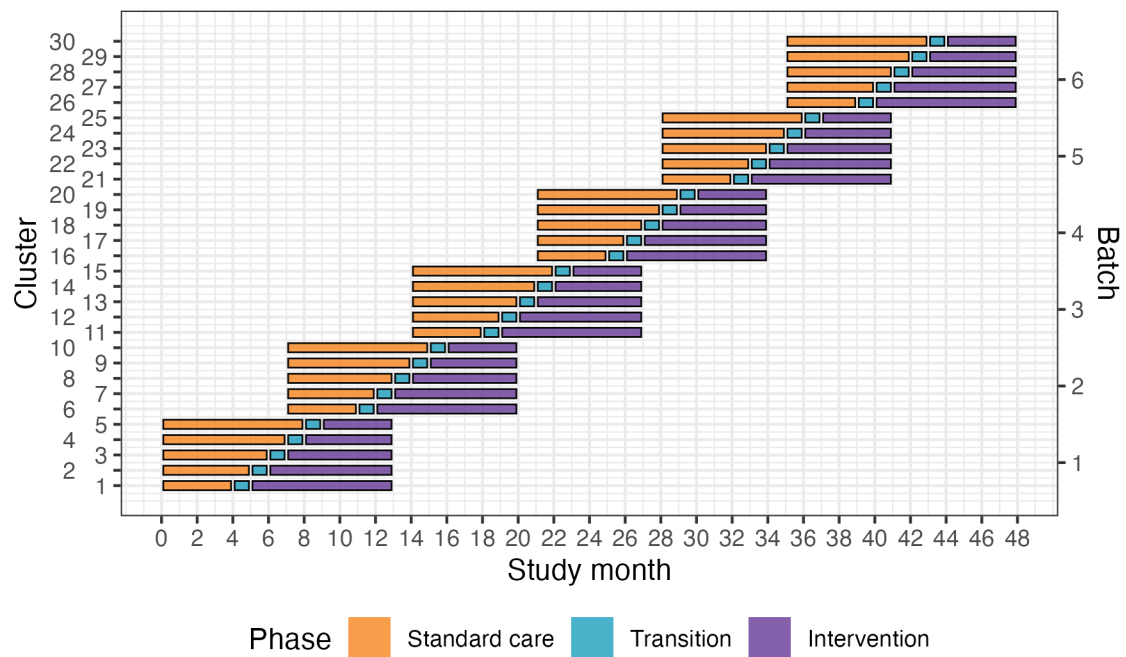


Figure 1: Trial design. Each line shows how long we include patients in a cluster, by batch and sequence. Clusters will be sequentially allocated to a batch based on when they enter the study. Within each batch clusters will then be randomised to an intervention implementation sequence.

Because certain secondary outcomes will be more burdensome to collect and because we anticipate larger effects on these outcomes, we will nest a staircase design within the main stepped-wedge design to measure these outcomes (see Figure 2)². The staircase design will include a random subset of patients presenting during the three months preceding the transition phase, and the three months following the transition phase.

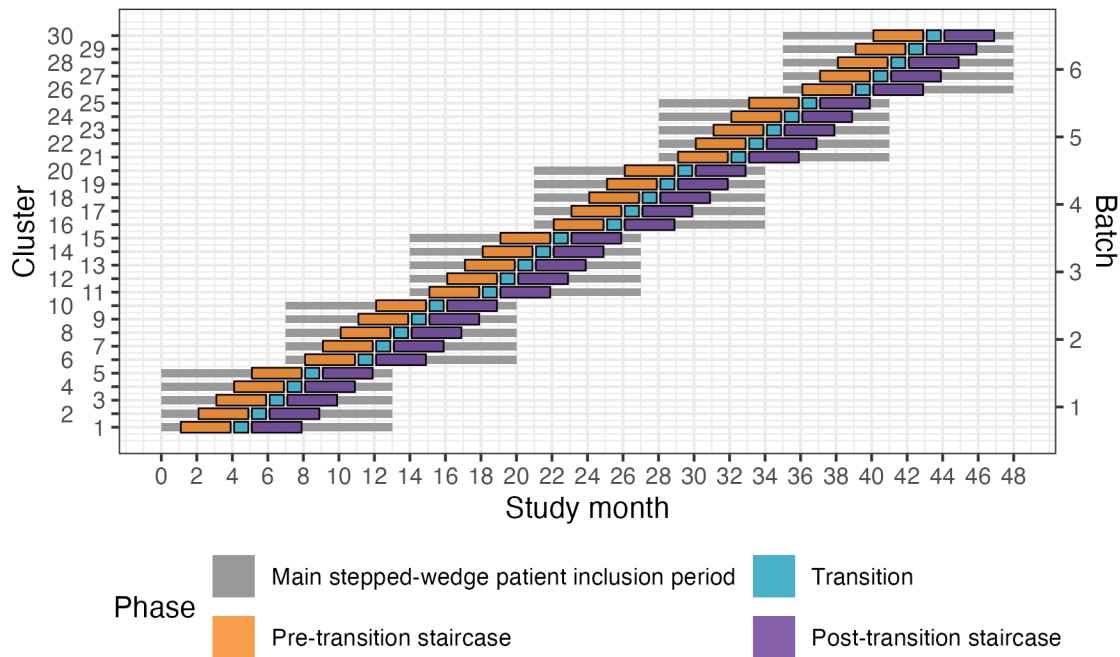


Figure 2: Nested staircase design. We include a random subset of patients presenting in the three months before and the three months after each cluster's transition phase.

4.2 Intervention and control treatment

The intervention will be ATLS® training. The control will be standard care, meaning no formal trauma life support training. We will train the physicians that initially resuscitate and provide trauma care during the first hour after patient arrival at the emergency department. These physicians can be casualty medical officers, surgical residents, or emergency medicine residents, depending on the setup at each participating centre. The training will occur during the transition phase in each cluster. Our experience from our pilot study is that study sites adhere to the training slot allotted to them through the trial, so we judge the risk of clusters implementing ATLS® before their randomised implementation sequence as very low.

We will train the number units of physicians needed to reach the required patient sample size, but estimate that this will require training an average of ten physicians per hospital, which on average should be mean that we can train one to two units per hospital. This is possible because many hospitals in India organise physicians staffing their emergency departments in units, and the physicians in the same unit

work together in the emergency department on the same days of the week. We will therefore collect data only on the days when these units work. The units selected to constitute a cluster from each hospital will be a convenience sample out of all eligible units in those hospitals.

Advanced Trauma Life Support® (ATLS®)³ is a proprietary 2.5 day course teaching a standardised approach to trauma patient care using the concepts of a primary and secondary survey. The programme was developed by the Committee of Trauma of the American College of Surgeons. The course includes initial treatment and resuscitation, triage and interfacility transfers. Learning is based on practical scenario-driven skill stations, lectures and includes a final performance proficiency evaluation. Physicians will be trained in an accredited ATLS® training facility in India. We will assess adherence to ATLS principles before and after implementing ATLS training.

Standard care varies across hospitals in India, but trauma patients are initially managed by casualty medical officers, surgical residents, or emergency medicine residents. They are mainly first- or second-year residents who resuscitate patients, perform interventions and refer patients for imaging or other investigations. Compared with other settings where a trauma team approach is adopted, nurses and other healthcare professionals are only involved to a limited extent during the initial management.

4.3 Randomisation and inclusion

We will enrol and assign clusters to batches once eligibility is confirmed and ethical approval is obtained. Each batch will include hospitals from different regions to optimise trial logistics and training delivery. Within each batch, clusters will be randomised to the intervention implementation sequences allocated to that batch.

We randomise clusters, stratified by batch. Within each batch we balance allocation on cluster size, defined as the monthly volume of eligible patients, using covariate-constrained randomisation (CCR). We chose CCR to improve balance between intervention conditions on this key design variable.

For example, in Batch 1, five clusters were randomised to one of five sequences using CCR. JM wrote an R program to enumerate all possible allocations ($n = 120$) and calculate a balance metric reflecting how evenly cluster size would be distributed between the two intervention conditions. For each allocation JM calculated the squared difference in anticipated cluster size between conditions; because there was only one covariate, this squared difference was the balance statistic.

We excluded the 10% of allocations with the greatest imbalance and kept the rest as the candidate set, then drew the final allocation at random from that set. Because of limitations in the algorithm, we calculated the balance score assuming a complete design, although the trial design includes transition periods. We will repeat this CCR procedure independently for each remaining batch.

JM generates the random allocation sequence for each batch and sends it to the Trial Team so they can begin logistical arrangements, including securing places on upcoming ATLS® training courses. We will conceal the allocation order from participating clusters for as long as is logistically feasible, because releasing physicians for training requires advance planning.

We select patients for the nested staircase outcomes separately from cluster randomisation, by stratified random sampling by shift; see Nested staircase design and secondary outcomes (Section 4.4.2).

4.4 Sample size calculations

We calculated the sample size during the trial design phase using the Shiny CRT Calculator⁴.

4.4.1 Main stepped-wedge design and primary outcome

With 30 clusters across 6 batches and a total sample size of 4320 our study has ~90% power across different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC) to detect a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS[®] training (see Figure 3 A)⁴. This 5% absolute reduction in mortality is a clinically meaningful and conservative estimate based on our updated systematic review and pilot study^{5,6}. We allowed for the clustered design, incorporated the one month transition period, and assumed: a discrete time decay correlation structure, an ICC of 0.02 (but considered sensitivity across the range 0.01-0.05), and a CAC of 0.9 (but considered sensitivity across the range 0.8-1.0), based on our pilot study and current guidance⁷⁻¹⁰. We included the CAC to allow for variation in clustering over time. We assume that each cluster will contribute approximately 12 observations per month to the analysis, but allowed for substantial variation in cluster sizes, based on our previous work. We also estimated the sample size required to detect a reduction in the primary outcome from 10% to 7.5%, which would require at least 30 observations per period and cluster (see Figure 3 B).

4.4.2 Nested staircase design and secondary outcomes

The secondary outcomes measured using the nested staircase design are adherence to ATLS[®] principles during initial patient resuscitation, quality of life, and disability. The power calculations presented here are intended only to inform the sample size required for their data collection within the nested staircase design. The expected effects of the intervention on these outcomes are an improvement in adherence from 50% during standard care to 70% after training¹¹, an increase in EQ-5D-5L health status from 70 during care to 75 after training¹², and finally a decrease in disability from a baseline value of 25 during standard care to 22.5 after training¹³. For quality of life and disability, these effects correspond to standardised effect sizes, expressed as Cohen's d, of 0.5. With 30 clusters in total, six per sequence, for a discrete time decay correlation structure, with ICCs of 0.01 to 0.15 and a CAC of 0.8, there is >80% power to detect these effects by including four patients in each cluster in each period. Accounting for loss to follow-up, we will include at least six patients per cluster per month. These patients will be a random subset of patients included in the primary outcome data collection during the staircase months. The random subset will be selected using stratified random sampling by shift, meaning that the timing of the clinical research coordinator's shift will be randomised to cover approximately eight hours during the morning, afternoon, or night shift. Adherence to ATLS[®] principles, quality of life, and disability will be measured in all patients included during these shifts. In each hospital, the number of shifts that will be randomised will be determined by the volume of patients included during the months preceding the staircase months.

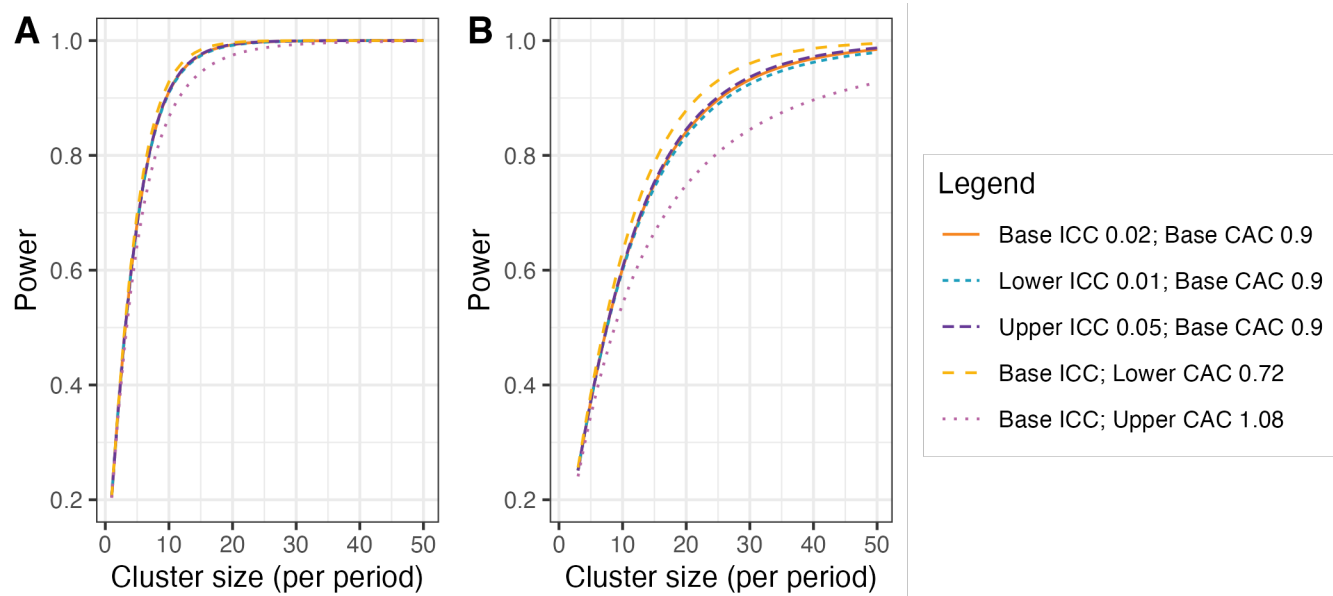


Figure 3: Power curves for different combinations of cluster autocorrelations (CAC) and intra-cluster correlations (ICC). **A**) Shows power curves assuming a reduction in the primary outcome of in-hospital mortality within 30 days from 20% under standard care to 15% after ATLS[®] training. **B**) Shows power curves assuming a reduction in the primary outcome from 10% under standard care to 7.5% after ATLS[®] training. Under this scenario, we would need to increase the sample size per month to around 30 observations to achieve 90% power under most combinations of CAC and ICC.

4.5 Interim analysis

There will be one interim analysis after half of the batches have completed the trial. The interim analysis will be assessed by the joint Trial Steering and Data Monitoring Committee. The purposes of this interim analysis will be to assess the trial's feasibility and recommend that the trial be stopped if it is not feasible (e.g. if hospitals fail to adhere to the randomisation schedule or if there are substantial missing data in outcomes), and to compare characteristics across intervention conditions to monitor for differential recruitment/ascertainment between the intervention and control groups. No efficacy stopping is planned because any estimates obtained at the time point of the interim analysis will be associated with major statistical uncertainties.

The Trial Team will prepare the interim analysis report for the joint Trial Steering and Data Monitoring Committee (SDMC), and the Trial Management Group will approve it before circulation. The report will follow the standard SDMC report template in the charter, except that it reports outcome data as completeness only, and will include:

- trial progress: numbers of clusters included and dropping out; numbers of potentially eligible patients screened, patients included, patients who did not consent to out-of-hospital follow-up, and patients lost to follow-up, overall and by intervention condition;
- adherence to the randomisation schedule and reasons for non-adherence;
- summary statistics of included clusters and patient participants by intervention condition;
- completeness of primary and secondary outcome data, overall and by intervention condition;
- ATLS® training delivery, including numbers of physicians trained in each hospital;
- protocol deviations and proposed protocol amendments;
- safety events, as defined for SDMC monitoring;
- relevant external evidence; and
- the SDMC recommendation.

We will not present or compare outcome values by intervention condition in the interim report, to avoid biasing the SDMC or Trial Management Group. The SDMC will use the interim report to advise whether the trial remains feasible and should continue as planned.

4.6 Final analysis

Primary and secondary outcomes will be analysed collectively in a single final analysis once all trial data are available.

5 Statistical principles

5.1 Statistical hypotheses

We will estimate the effect of ATLS® training on 30-day in-hospital mortality compared with standard care, as an odds ratio (OR) and as an absolute risk difference (ARD). An $OR < 1$ or $ARD < 0$ indicates

lower mortality among patients treated in hospitals after ATLS® implementation than under standard care. Based on our pilot study and systematic review of the literature, we expect the intervention to reduce mortality.

5.2 Levels of statistical significance and confidence

We will use a two-sided significance level of 0.05 for all analyses, and we will report 95% confidence intervals (CI) for all estimates.

5.3 Type I error control

We will not adjust for multiple testing because we do not regard any secondary outcome as singularly more important. We will interpret the secondary outcomes together and not over-emphasise a single statistically significant result when the others appear unchanged.

5.4 Descriptive summaries

In every descriptive table in this plan we summarise discrete variables as frequencies and percentages, and continuous and count variables as medians and interquartile ranges (Q1–Q3); where a variable fits neither, we use the summary statistic given in the patient characteristics tables. We report missing values for each variable, and we do not adjust any of these summaries for clustering. Where a table has a key and an all version, we present the key version in the results section and the all version as a supplementary table.

5.5 Adherence and protocol deviations

5.5.1 Adherence to the intervention

Adherence to the intervention is defined at the cluster and individual levels and is distinct from the secondary outcome adherence to ATLS® principles during resuscitation, see Analysis of secondary outcomes (Section 9.1.1).

At the cluster level, adherence means that ATLS® training is delivered during the scheduled transition period for that cluster's randomised sequence, and that the cluster does not implement ATLS® training before its scheduled transition. We will assess cluster level adherence using the training log kept by the trial team. We will summarise cluster-level adherence descriptively, including the number and proportion of clusters that completed training as scheduled and the reasons for any non-adherence to the randomisation schedule. Adherence to the randomisation schedule is also reviewed at the interim analysis, see Interim analysis (Section 4.5).

At the individual level, exposure to the intervention is defined by the cluster's scheduled transition date, and patients are classified by the period in which they arrived at the emergency department, see Analysis sets (Section 8.3). The primary analysis includes eligible patients irrespective of whether care was delivered by a trained physician.

Process adherence to ATLS® principles during initial resuscitation is a nested-staircase secondary outcome and is defined and analysed in Outcomes (Section 7.1) and Analysis of secondary outcomes (Section 9.1.1); it is not used to exclude patients from the primary analysis.

5.5.2 Protocol deviations

A protocol deviation is any departure from the approved trial protocol, ICH-GCP, or applicable regulations. Consistent with the protocol, we will report to the sponsor without delay, for assessment, any serious breach or deviation that significantly affects, or is likely to affect, participant safety or integrity or the reliability of trial data. We will document minor deviations — those that do not affect participant safety or integrity and do not materially affect the scientific value of the trial — in the trial records of the principal investigator and the sponsor.

We will summarise protocol deviations descriptively for the final analysis and review them at the interim analysis, overall and by intervention condition where informative, giving counts by type (cluster-level versus individual-level, and serious versus minor where classified) and narrative reasons. We will not exclude clusters or patients from the primary analysis because of a deviation; the primary analysis still uses scheduled intervention exposure and the outcome data available under the missing-data plan, as specified under Analysis sets (Section 8.3).

6 Trial sample

6.1 Eligibility criteria

Our trial has eligibility criteria at the cluster and patient participant levels. We summarise them below; the protocol gives full details.

6.1.1 Cluster eligibility criteria

6.1.1.1 Inclusion criteria

Clusters must meet the following criteria:

- admit or refer/transfer for admission at least 400 patients with trauma per year or 35 patients with trauma per month for at least the last six months;
- provide surgical and orthopaedic emergency services around the clock; and
- have at most 25% of physicians providing initial trauma care trained in a formalised trauma life support training programme, like ATLS® or Primary Trauma Care (PTC).

Additionally, the units to be trained within the cluster must meet the following criteria:

- admits or refers/transfers for admission at least 12 patients with trauma per month for at least the last six months; and
- no more than 25% of physicians providing initial trauma care trained in a formalised trauma life support training programme.

6.1.1.2 Exclusion criteria

Hospitals are excluded if they meet any of the following criteria:

- implement a formalised trauma life support training programme¹ during the trial period; or
- plan to implement or implement other major interventions² that affects trauma care during the trial period.

6.1.2 Patient participant eligibility criteria

6.1.2.1 Inclusion criteria

Patients participants must meet the following criteria:

- age of at least 15 years;
- trauma occurred less than 48 hours before arrival at the hospital;
- present to the emergency department of participating hospitals, with a history of trauma defined as having any of the reasons listed in the International Classification of Diseases chapter XX as the reason for presenting;
- admitted, or died between arrival at the hospital and admission, or referred/transferred from the emergency department of a participating hospital to another hospital for admission; and
- managed by a participating unit in the emergency department.

6.1.2.2 Exclusion criteria

Patients participants are excluded if they meet the following criteria:

- present with isolated closed extremity fracture; or
- are directly admitted to a ward without being seen by a physician in the emergency department.

6.2 Participant flow

We will use CONSORT diagrams to display the flow of clusters and patients through the main batched stepped-wedge trial, reported as two figures in the main text, in line with recent batched stepped-wedge practice¹⁴: a cluster-level diagram (Figure 4) and a patient-level diagram (Figure 5). For each sequence, the diagrams will report numbers randomly assigned or included, losses and exclusions after randomisation (with reasons), and numbers analysed for the primary outcome. The patient diagram will also summarise flow before and after ATLS® training. Flow for nested-staircase secondary outcomes (adherence to ATLS® principles, EQ-5D-5L, and WHODAS 2.0) will be reported in a separate figure in the supplementary material (Figure 6), showing patients in the staircase periods, losses and exclusions, and numbers analysed, by sequence, before and after ATLS® training, and overall. We will present

¹These include but are not limited to the National Emergency Life Support (NELS) programme, the Basic Trauma Life Support (BTLS) programme, the Pre-Hospital Trauma Life Support (PHTLS) programme, the Trauma Nursing Core Course (TNCC) and the Advanced Trauma Care for Nurses (ATCN) programme.

²These include but are not limited to implementing of a trauma team approach, opening a trauma centre and implementing a trauma quality improvement programme.

cluster-level summaries of the intervention effect. We will report the study according to the CONSORT guidelines for stepped-wedge randomised trials¹⁵.

6.3 Baseline characteristics

6.3.1 Cluster characteristics

We will describe cluster characteristics including state and size using frequencies and percentages for discrete variables and medians and interquartile ranges (Q1-Q3) for continuous variables. State categories depend on the final set of participating clusters. We will present these characteristics in the main text. See Table 2 for details.

6.3.2 Patient characteristics

We will describe patient characteristics among all eligible patients over the entire trial period, stratified by before and after ATLS training and overall, using frequencies and percentages for discrete variables and medians and interquartile ranges (Q1-Q3) for continuous variables. Missing values will be reported for each characteristic. We will not adjust for clustering when presenting baseline characteristics (i.e. non-outcome variables). We will present key characteristics in the main text, see Table 3, and present all characteristics in the supplementary material, see Table 4.

7 Outcomes

7.1 Outcomes

A summary of all outcomes, analysis sets, effect measures, and handling of intercurrent events is provided in Table 1. The strategies named in the last column of that table are defined under Strategies for handling intercurrent events (Section 8.1.1).

7.1.1 Primary outcome

The primary outcome will be in-hospital mortality within 30 days of arrival at the emergency department. We define this outcome as any death occurring in the hospital during the initial admission, up to 30 days post arrival at the emergency department. Clinical research coordinators will extract information on death from patient hospital records. If the patient has been transferred to another hospital, the clinical research coordinators will collect data on this outcome by calling the patient or a patient representative, or by contacting the hospital to which the patient was transferred. Data on this outcome will be collected continuously during the trial.

7.1.2 Secondary outcomes

7.1.2.1 Collected using the main stepped-wedge design

- All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department. In contrast to the primary outcome, all-cause mortality includes all deaths regardless of setting. Data on this outcome will be collected in the same way as for the primary outcome.
- Length of emergency department stay. Data on this outcome will be collected from patient hospital records.
- Length of hospital stay. Data on this outcome will be collected from patient hospital records.
- Intensive care unit admission. Data on this outcome will be collected from patient hospital records.
- Length of intensive care unit stay. Data on this outcome will be collected from patient hospital records.
- Return to work at 30 days and three months after arrival at the emergency department. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged.

7.1.2.2 Collected using the nested staircase design

- Adherence to ATLS® principles during initial patient resuscitation, up to one hour after the physician has first seen the patient. This assessment will be done using a 14 item checklist covering the key steps of the ATLS® primary survey, which was modelled based on previous work on ATLS® adherence¹⁶. We will consider completion of all 14 steps as 100% adherence. The clinical research coordinators collecting the data will be trained by the trial team to do this, prior to the start of the trial. We will collect this data by observing the care of a random sample of patients. The sampling will be designed as a nested staircase design.
- Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, measured by the official and validated translations of the EQ5D5L. This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. We will collect this data using a nested staircase design.
- Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, assessed using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0). This outcome is only relevant to alive patients. Data on this outcome will be collected in person if the patient is still in hospital, or by phone if the patient has been discharged. This data will also be collected using a nested staircase design.

Table 1: Summary of outcomes, analysis sets, effect measures, and handling of intercurrent events. The strategies named in the last column are defined in the section Strategies for handling intercurrent events.

Outcome	Design component		Data type	Analysis set	Main analysis model		Effect measure	Potential intercurrent events	Strategy for handling intercurrent events
Primary outcome: In-hospital mortality within 30 days of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Primary analysis set	Mixed-effects binomial (logit, identity)	binomial (logit, OR; ARD	OR; ARD	Loss to follow-up; withdrawal	Missing data
All-cause mortality within 24 hours of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Main stepped-wedge, outcome observed	Mixed-effects binomial (logit, identity)	binomial (logit, OR; ARD	OR; ARD	Loss to follow-up; withdrawal	Treatment policy (any setting); missing data otherwise
All-cause mortality within 30 days of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Main stepped-wedge, outcome observed	Mixed-effects binomial (logit, identity)	binomial (logit, OR; ARD	OR; ARD	Loss to follow-up; withdrawal	Treatment policy (any setting); missing data otherwise
All-cause mortality within three months of arrival at the emergency department	Main wedge	stepped-	Dichotomous	Main stepped-wedge, outcome observed	Mixed-effects binomial (logit, identity)	binomial (logit, OR; ARD	OR; ARD	Loss to follow-up; withdrawal	Treatment policy (any setting); missing data otherwise
Length of emergency department stay	Main wedge	stepped-	Count (hours)	Main stepped-wedge, outcome observed	Mixed-effects negative binomial (log)	binomial (logit, OR; ARD	Rate ratio	Death; transfer	Observed stay truncated by death or transfer
Length of hospital stay	Main wedge	stepped-	Count (days)	Main stepped-wedge, outcome observed	Mixed-effects negative binomial (log)	binomial (logit, OR; ARD	Rate ratio	Death; transfer	Observed stay truncated by death or transfer
Intensive care unit (ICU) admission	Main wedge	stepped-	Dichotomous	Main stepped-wedge, outcome observed	Mixed-effects binomial (logit, identity)	binomial (logit, OR; ARD	OR; ARD	Death; transfer	Classified by whether ICU admission occurred first
Length of intensive care unit stay	Main wedge	stepped-	Count (days)	ICU admissions	Mixed-effects negative binomial (log)	binomial (logit, OR; ARD	Rate ratio	Death; transfer	Observed stay truncated by death or transfer
Return to work at 30 days after arrival at the emergency department	Main wedge	stepped-	Dichotomous	Alive at 30 days, outcome relevant	Mixed-effects binomial (logit, identity)	binomial (logit, OR; ARD	OR; ARD	Death; loss to follow-up; withdrawal	While-alive; missing data otherwise
Return to work at three months after arrival at the emergency department	Main wedge	stepped-	Dichotomous	Alive at 3 months, outcome relevant	Mixed-effects binomial (logit, identity)	binomial (logit, OR; ARD	OR; ARD	Death; loss to follow-up; withdrawal	While-alive; missing data otherwise
Adherence to ATLS® principles during initial patient resuscitation	Nested staircase		Proportion	Observed nested sample	Mixed-effects beta (logit)	Logit-scale difference		Incomplete observation	Available-case in the observed nested sample
Quality of life within seven days of discharge	Nested staircase		Ordinal EQ-5D-5L; VAS	Alive, nested sample	Cumulative logit; linear mixed	COR; mean difference		Death; loss to follow-up; withdrawal	While-alive; missing data otherwise
Quality of life at 30 days after arrival at the emergency department	Nested staircase		Ordinal EQ-5D-5L; VAS	Alive at 30 days, nested sample	Cumulative logit; linear mixed	COR; mean difference		Death; loss to follow-up; withdrawal	While-alive; missing data otherwise
Quality of life at three months after arrival at the emergency department	Nested staircase		Ordinal EQ-5D-5L; VAS	Alive at 3 months, nested sample	Cumulative logit; linear mixed	COR; mean difference		Death; loss to follow-up; withdrawal	While-alive; missing data otherwise
Disability within seven days of discharge	Nested staircase		Ordinal WHO-DAS 2.0; summary score	Alive, nested sample	Cumulative logit; linear mixed	COR; mean difference		Death; loss to follow-up; withdrawal	While-alive; missing data otherwise

(continued)

Outcome	Design component	Data type	Analysis set	Main analysis model	Effect measure	Potential events	intercurrent	Strategy for handling intercurrent events
Disability at 30 days after arrival at the emergency department	Nested staircase	Ordinal WHO-DAS 2.0; summary score	Alive at 30 days, nested sample	Cumulative logit; linear mixed	COR; mean difference	Death; loss to follow-up; withdrawal		While-alive; missing data otherwise
Disability at three months after arrival at the emergency department	Nested staircase	Ordinal WHO-DAS 2.0; summary score	Alive at 3 months, nested sample	Cumulative logit; linear mixed	COR; mean difference	Death; loss to follow-up; withdrawal		While-alive; missing data otherwise

8 Intercurrent events, populations and summary measures

8.1 Intercurrent events

Intercurrent events are events occurring after intervention exposure that affect interpretation or occurrence of an outcome. Anticipated events and the strategy used for each outcome are summarised in Table 1. Frequencies will be summarised for each outcome as counts and percentages of patients with the event before assessment of that outcome, stratified by before and after ATLS training and overall. We will not adjust for clustering in these summaries. Denominators are the primary analysis set or the nested sample, before the outcome-specific restrictions under Analysis sets (Section 8.3), so that events that remove a patient from an analysis set are still counted. We will present these summaries in the supplementary material. See Table 5.

At the cluster level, late or early transition relative to the scheduled date, incomplete delivery of ATLS® training, and cluster dropout are treated as protocol or adherence issues, as described under Adherence and protocol deviations (Section 5.5), rather than as outcome-specific intercurrent events; exposure for analysis remains as defined under Analysis sets (Section 8.3).

At the individual level, patients cannot withdraw from the intervention. For length of emergency department, hospital, and intensive care unit (ICU) stay, the observed stay is the completed duration to exit; death and transfer truncate it, and we do not treat either as censoring. For ICU admission we record whether the patient was admitted to ICU before death or transfer; patients who die or are transferred without being admitted to ICU count as not admitted.

- **Transfer to another hospital.** Relevant to length-of-stay and ICU outcomes. It is not an intercurrent event for mortality outcomes, because the primary outcome includes in-hospital deaths after transfer and all-cause mortality counts deaths regardless of care setting.
- **Death.** Relevant to length-of-stay and ICU outcomes and to survivor-only outcomes. As an intercurrent event, death is death before assessment of the outcome in question; later death is not an intercurrent event for earlier outcomes. Observed length of stay is the completed duration to exit; death truncates the observed stay and is not treated as censoring. ICU admission is classified according to whether admission occurred before death. Quality of life, disability, and return to work are defined only among patients alive at the relevant follow-up (while-alive restriction). Those analyses therefore target effects in survivors at that time point; mortality is analysed separately.
- **Withdrawal of consent for non-routine data collection.** Relevant only to outcomes that require opt-in follow-up: all-cause mortality after the initial admission (when not available from records), return to work, quality of life, and disability. It is also relevant to the primary outcome when death after transfer cannot be ascertained from records and requires contact with the patient, a representative, or the receiving hospital. Withdrawal is not listed for routinely recorded in-hospital outcomes (length of stay and ICU admission) or for adherence, which are collected under opt-out consent. Participants who withdraw contribute observed non-routine outcomes up to withdrawal unless they also request removal of already collected data; subsequent missing values are handled under the missing-data plan, see Missing data (Section 9.3).

- **Loss to follow-up.** Relevant to the same contact-dependent outcomes as withdrawal. These events create missing outcome data and are handled under the missing-data plan.
- **Incomplete observation of resuscitation.** Relevant only to adherence. We score the checklist by observing the initial resuscitation up to one hour after the physician first sees the patient. Observation is incomplete when we do not observe that whole window — for example when the observer could not complete the 14-item checklist because care ended, the patient left the emergency department, or observation was interrupted. We exclude unobserved and incompletely observed resuscitations from the adherence analysis (available-case analysis of the observed nested sample), and we treat death during the observation window the same way.

Return to work is further restricted to patients for whom the outcome is relevant, as defined under Analysis sets (Section 8.3); that restriction defines the analysis set and is not an intercurrent event.

8.1.1 Strategies for handling intercurrent events

The strategies to handle intercurrent events are defined as follows, in the order in which they first appear in Table 1. Where two strategies are listed for an outcome, the second applies to any event not covered by the first.

- **Missing data.** The outcome is not observed and no value is constructed for it. Such observations are excluded from the analysis of that outcome under the available-case approach, or imputed under the multiple imputation approach, as specified under Missing data (Section 9.3).
- **Treatment policy (any setting).** The outcome is counted as recorded irrespective of where care was delivered after the patient left the randomised cluster. All-cause mortality therefore counts deaths in any setting, including after transfer or discharge.
- **Observed stay truncated by death or transfer.** The length of stay is the completed duration until the patient leaves the unit, including when the patient leaves by dying or being transferred. These stays enter the analysis at their observed value and are not treated as censored.
- **Classified by whether ICU admission occurred first.** Patients who die or are transferred count as admitted to intensive care only if the admission occurred before that event. The outcome is therefore defined for every patient in the analysis set.
- **While-alive.** The outcome is defined only among patients alive at the assessment time point. The estimated effect applies to survivors at that time point, and mortality is analysed separately.
- **Available-case in the observed nested sample.** The outcome is analysed among the resuscitations that were fully observed. Unobserved and incompletely observed resuscitations are excluded.

Where a while-alive strategy is used, we assume that, conditional on the analysis model and design factors, death before the outcome assessment does not induce selection bias beyond what is addressed by analysing mortality separately; sensitivity of conclusions will rely on joint interpretation of mortality and survivor outcomes. Assumptions underlying each strategy will be considered when interpreting results and when checking model assumptions, see Analysis methods (Section 9).

8.2 Population

A population is the group of clusters and patients to which an estimated effect applies. An analysis set is the subset of a population whose observations enter a given analysis, as defined under Analysis sets (Section 8.3). The treatment effects estimated in this plan are defined for the trial's eligible clusters and eligible patient participants, as specified under Eligibility criteria (Section 6.1).

- **Clusters.** We estimate effects for participating hospitals (clusters) that meet the cluster eligibility criteria and that we randomise within the batched stepped-wedge design. Units within a hospital are part of the same cluster; we do not define the estimands separately by unit.
- **Main stepped-wedge patient population.** For the primary outcome, all-cause mortality secondary outcomes, length of emergency department and hospital stay, and ICU admission, the target population is eligible trauma patients included in the main stepped-wedge trial during non-transition periods. For length of ICU stay the population is those patients who are admitted to ICU.
- **Survivor populations.** We target quality of life and disability effects among patients in the nested staircase sample who are alive at the relevant follow-up time point, and return-to-work effects among patients in the main stepped-wedge trial who are alive at the relevant follow-up and for whom return to work is relevant.
- **Nested adherence population.** We target the adherence-to-ATLS®-principles effect among the patients we randomly select for observation in the nested staircase design during the observation window.

8.3 Analysis sets

An observation is one presentation by an eligible patient to the emergency department of a participating cluster, see Eligibility criteria (Section 6.1). Each observation belongs to the cluster the patient presented to and to the period in which the patient arrived at the emergency department. Every randomised cluster contributes observations, and a cluster that leaves the trial early contributes the periods it completed. Observations from a cluster's transition period are excluded from all analyses.

An observation is classified as exposed to the intervention if the patient arrived after the period on which the cluster was scheduled to transition, and as standard care if the patient arrived before that period. Exposure does not depend on whether a trained physician delivered the care, or on when the cluster was in fact trained, see Adherence and protocol deviations (Section 5.5). If a cluster transitions more than four weeks from its scheduled date, we will repeat the primary analysis with exposure defined by the actual date of transition, see Exposure defined by the actual date of transition (Section 9.4.1).

The observations for the patients selected for nested staircase data collection form the nested sample. These patients are drawn by stratified random sampling by shift during the three months before and the three months after the transition period, see Nested staircase design and secondary outcomes (Section 4.4.2).

Table 1 names an analysis set for each outcome. The sets are defined as follows, in the order in which they first appear in that table.

- **Primary analysis set.** All observations, conditioned on the missing data strategy used. This set is used for the primary outcome and for the secondary outcomes collected with the main stepped-wedge design.
- **ICU admissions.** The observations for patients in the primary analysis set who were admitted to intensive care.
- **Alive at 30 days and working at the time of inclusion.** The observations for patients who were alive 30 days after arrival at the emergency department and who were working at the time of inclusion. Working means paid or unpaid work, studying or keeping house.
- **Alive at 3 months and working at the time of inclusion.** The same restriction applied three months after arrival at the emergency department.
- **Observed nested sample.** The observations in the nested sample for which the resuscitation was observed in full.
- **Alive, nested sample.** The observations in the nested sample for patients who were alive when the outcome was assessed.
- **Alive at 30 days, nested sample.** The observations in the nested sample for patients who were alive 30 days after arrival at the emergency department.
- **Alive at 3 months, nested sample.** The observations in the nested sample for patients who were alive three months after arrival at the emergency department.

An analysis includes the observations in the analysis set for that outcome whose outcome value is observed. Under multiple imputation, observations with a missing outcome are retained and the value is imputed, see Missing data (Section 9.3). Adjusted analyses under the available-case approach include only observations with complete data on the adjustment covariates.

8.4 Effect measures

Outcomes are measured on individual patients. Descriptive summaries of primary and secondary outcomes will be presented stratified by before and after ATLS® training and overall, using frequencies and percentages for dichotomous and categorical outcomes, medians and interquartile ranges (Q1–Q3) for continuous and count outcomes, and the same summary statistics as the patient characteristics tables otherwise. Missing values will be reported for each outcome. These descriptive summaries are unadjusted for clustering (patient-level summaries). We will present key outcome summaries in the main text, and all outcome summaries in the supplementary material. See Table 6 for key outcomes and Table 7 for all outcomes.

Table 1 lists the comparative effect measures by outcome; we estimate them from the mixed-effects models in Analysis methods (Section 9). These models give cluster-specific (conditional) estimates of the intervention effect for patient-level outcomes, not marginal (population-averaged) estimates. We target the patient-level average among eligible patients in participating clusters, rather than an equal-weight average of cluster-level summaries. The types of effect measures are as follows:

- **Dichotomous outcomes:** odds ratio (logit link) and absolute risk difference (identity link), where both links are used.
- **Count (length-of-stay) outcomes:** rate ratio from a negative binomial model with log link.

- **Ordinal outcomes (EQ-5D-5L domains and WHODAS domain scores):** cumulative odds ratio from a cumulative logit model.
- **Continuous outcomes (EQ-5D-5L VAS and WHODAS summary score):** mean difference from a linear mixed model.
- **Adherence proportion:** difference on the logit scale from a beta mixed model.

Shell tables of the planned comparative results are given at the end of this plan (Table 8, Table 9, and Table 10).

9 Analysis methods

Primary and secondary outcomes will be analysed with mixed-effects regression models that account for clustering at the hospital level and target cluster-specific intervention effects. The link function and outcome distribution are matched to the data type of each outcome, and the model for each outcome is specified in the subsections below.

Randomisation is at the hospital level, and all participating units within a hospital belong to the same cluster. The models therefore include a random effect at the hospital level and no separate random effect for unit within hospital.

Unless stated otherwise, the estimation and reporting specifications below apply to all analyses in this section. Estimation will be by maximum likelihood, with a Laplace approximation used to integrate out the random effects. The specific R package and numerical implementation are not prespecified, see Statistical software (Section 9.8).

Standard errors will be model-based. Confidence intervals will be Wald-type. To correct potential inflation of the type I error rate due to the small number of clusters, a small-sample correction will be applied. Our primary method will be the Kenward–Roger correction¹⁷, but this choice may be updated based on methodological developments available closer to the end of patient inclusion. The correction applied may also differ for outcomes collected via the complete stepped-wedge design and the incomplete nested staircase designs.

Model diagnostics will be used to assess distributional assumptions and model convergence.

9.1 Analysis of the primary outcome

The primary outcome is in-hospital mortality within 30 days of arrival at the emergency department, which we will analyse as a dichotomous variable. We will estimate the odds ratio (OR) and the absolute risk difference (ARD) of death under ATLS[®] compared with standard care.

We will use a mixed-effects binomial regression model to estimate the effect of ATLS[®] on mortality. We will estimate the OR using the logit link and the ARD using the identity link. These models account for clustering at the hospital level and give cluster-specific estimates of the intervention effect. Our primary estimand is a common intervention effect across periods of ATLS[®] exposure — a time-averaged effect under a constant treatment-effect assumption — rather than separate period-specific effects

after transition. For the primary analysis we will fit a prespecified sequence of models of decreasing complexity and report the most complex one that converges, moving to simpler models only if we meet convergence or numerical stability problems. We will then run the sensitivity analyses specified below. If the binomial model with the logit link converges but the corresponding model with the identity link does not, then we will report only an OR.

All models in this sequence will include a fixed effect for intervention exposure and adjust for time, although the parametrisation of time and the random effects structure may be simplified.

The model sequence is defined as follows:

1. **Primary model:** a model including fixed effects for period, modelled as a categorical variable with separate period effects for each batch, and a fixed effect for intervention exposure. Clustering will be accounted for by including a random intercept for cluster and a random cluster-by-period effect, both assumed to follow a normal distribution (Equation 1).

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (1)$$

2. **Simplified random effects structure:** a model including fixed effects for period and intervention exposure and a random intercept for cluster only, thereby removing the random cluster-by-period effect (Equation 2).

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} \quad (2)$$

3. **Model with shared period effects:** a model in which fixed period effects are shared across batches, with time adjusted for using a common period effect or calendar time modelled directly (e.g., as a continuous covariate or using splines). A random intercept for cluster and a random cluster-by-period effect are retained; clusters remain indexed by batch, but the fixed period effects are common across batches (Equation 3).

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (3)$$

4. **Combined simplified model:** a model that includes both a simplified random effects structure (random intercept for cluster only) and shared period effects. This represents the most parsimonious mixed-effects model considered acceptable for the primary analysis (Equation 4).

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} \quad (4)$$

Where:

- $g(\cdot)$: the link function, specified as either the logit link, $g(p) = \log\{p/(1-p)\}$, or the identity link, $g(p) = p$, depending on the estimand of interest.

- θ : **the intervention effect**, represented as a fixed effect of intervention exposure (the primary estimand); under the logit link this represents the log-odds ratio for death associated with ATLS® exposure compared with standard care, and under the identity link it represents the absolute risk difference.
- $\Pr(Y_{bkti} = 1)$: the probability of death for patient $i = 1, \dots, m_{bkt}$ in cluster $k = 1, \dots, 5$ during period $t = 1, \dots, 12$ in batch $b = 1, \dots, 6$, giving 30 clusters in total.
- m_{bkt} : the number of patients in cluster k during period t in batch b .
- μ : the intercept, representing the value of the linear predictor for the reference levels of all fixed effects, with random effects equal to zero.
- β_t : the fixed effect of period t , shared across batches.
- β_{bt} : the fixed effect of period t in batch b , accounting for a separate period effect for each batch, resulting in a total of $6 \times (12)$ period effects.
- X_{bkt} : the intervention exposure indicator for cluster k during period t in batch b , with $X_{bkt} = 1$ for ATLS® and $X_{bkt} = 0$ for standard care.
- α_{bk} : a normally distributed random effect for cluster k in batch b , representing between-cluster (hospital-level) variability in the outcome.
- γ_{bkt} : a normally distributed random effect for cluster k in period t in batch b , representing within-cluster variation over time and capturing cluster-specific deviations from the average period effect.

The distributional assumptions for the random effects are specified in Equation 5.

$$\begin{aligned}\alpha_{bk} &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\alpha}^2), \\ \gamma_{bkt} &\stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_{\gamma}^2), \\ \alpha_{bk} &\perp \gamma_{bkt}.\end{aligned}\tag{5}$$

9.1.1 Analysis of secondary outcomes

Unless otherwise stated, secondary outcomes will be analysed using the model specification selected for the primary analysis under Analysis of the primary outcome (Section 9.1), with the link function and outcome distribution adapted to the scale of the outcome. The period parametrisation and random effects structure of that specification are carried over, so the sequence of models of decreasing complexity is not traversed again for each secondary outcome. If the specification selected for the primary analysis does not converge for a secondary outcome, we will simplify the structure for that outcome by continuing down the same sequence, and we will report which specification was used for each outcome. For outcomes collected within the nested staircase design, period effects are defined on the global study timeline rather than separately within each batch, as described in the subsections below; the random effects structure still follows the primary analysis.

Effect measures are odds ratios and absolute risk differences for binary outcomes, rate ratios for count (length-of-stay) outcomes, cumulative odds ratios for ordinal outcomes, mean differences for continuous outcomes, and a logit-scale difference for the adherence proportion. These models provide cluster-specific estimates of the intervention effect for patient-level outcomes. For cumulative logit models, diagnostics will also assess the proportional-odds assumption.

9.1.1.1 All cause mortality within 24 hours, 30 days and three months of arrival at the emergency department

These outcomes will be analysed as dichotomous variables using the mixed-effects binomial model selected for the primary analysis, with the logit and identity links, to estimate odds ratios and absolute risk differences.

9.1.2 Quality of life within seven days of discharge, and at 30 days and three months of arrival at the emergency department, among survivors

We measure quality of life within the nested staircase design, using the official and validated translations of the EQ-5D-5L. It yields two outcome types: five ordinal domain scores (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression), each rated on a Likert scale from 1 to 5, and a visual analogue scale (VAS) ranging from 0 to 100.

For each of the five ordinal domains, we will use a mixed-effects cumulative logit (proportional odds) model:

$$\text{logit}\{\Pr(Y_{bkti} \leq j)\} = \mu_j + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (6)$$

Where μ_j is the intercept for the j th category.

The VAS will be analysed as a continuous outcome using a linear mixed-effects model:

$$\text{VAS}_{bkti} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + \epsilon_{bkti}, \quad \epsilon_{bkti} \sim \mathcal{N}(0, \sigma^2) \quad (7)$$

9.1.3 Disability within seven days of discharge, and at 30 days and three months of arrival at the emergency department, among survivors

We measure disability within the nested staircase design using the WHO Disability Assessment Schedule 2.0 (WHODAS 2.0)¹⁸, which assesses six domains of functioning: cognition, mobility, self-care, getting along, life activities, and participation. Each domain is rated on a Likert scale from 1 to 5, with 1 indicating no difficulties and 5 indicating extreme difficulties. We will analyse each domain separately using a mixed-effects cumulative logit model (Equation 6). We will also calculate a WHODAS 2.0 summary score using the “complex scoring” method: we sum the item scores within each of the six domains, sum the six domain scores, and transform the total to a 0–100 scale. We will analyse the summary score using a linear mixed-effects model (Equation 7).

9.1.4 Return to work at 30 days and three months after arrival at the emergency department, among survivors

We will analyse return to work as a dichotomous outcome using the mixed-effects binomial model selected for the primary analysis, with the logit and identity links, to estimate odds ratios and absolute risk differences.

9.1.5 Length of emergency department stay

We will analyse length of emergency department stay as a count outcome: the number of hours from arrival at the emergency department (ED) until ED exit. Exit may occur via ward admission, ICU admission, transfer to another hospital, death, or discharge home; all of these routes contribute the observed ED duration. We will estimate the effect of ATLS® training on mean ED length of stay. We analyse mortality separately as a primary and secondary outcome rather than as a competing event.

The model is a mixed-effects negative binomial regression with a log link, including a fixed effect for intervention exposure and the period parametrisation and random effects structure selected for the primary analysis. Equation 8 shows this model with batch-specific categorical period effects, a random intercept for cluster, and a random cluster-by-period effect; if a simpler structure was selected for the primary analysis, that structure is used here instead. If the model does not converge for this outcome, we will simplify it further by continuing down the same sequence. The intervention effect θ is the log rate ratio for ED length of stay associated with ATLS® exposure compared with standard care; we will also report the corresponding multiplicative effect on the mean count (rate ratio).

$$\log\{\mathbb{E}(Y_{bkti})\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (8)$$

Where Y_{bkti} is the length-of-stay count for patient i in cluster k during period t in batch b , and all other terms are defined as in the primary analysis model. The negative binomial model also estimates a dispersion parameter.

9.1.5.1 Length of hospital stay

Length of hospital stay will be analysed as a count outcome: the number of days from arrival at the emergency department to hospital discharge (or death in hospital). The analysis uses the same negative binomial specification as for ED length of stay (Equation 8), with the same handling of death and transfer.

9.1.6 Intensive care unit admission

Intensive care unit admission will be analysed as a dichotomous outcome using the mixed-effects binomial model selected for the primary analysis, with the logit and identity links, to estimate odds ratios and absolute risk differences. Patients who die or are transferred without ICU admission will be classified according to whether ICU admission occurred before that event.

9.1.6.1 Length of intensive care unit stay

Length of intensive care unit stay will be analysed as a count outcome among patients admitted to ICU: the number of days from ICU admission to ICU discharge (or death in ICU). The analysis uses the same negative binomial specification as for the other length-of-stay outcomes (Equation 8), with the same handling of death and transfer.

9.1.7 Adherence to ATLS® principles during initial patient resuscitation

We assess adherence to ATLS® principles within the nested staircase design, in both the standard care and the ATLS® periods, as the percentage of completed items on a 14-item checklist that covers the key steps of the ATLS® primary survey, where all 14 steps completed is 100% adherence. We will rescale the percentage to a proportion on the 0–1 scale and analyse it with a mixed-effects beta regression model with a logit link:

$$\text{logit}\{\mathbb{E}(Y_{bkti})\} = \mu + \beta_t + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (9)$$

Where $\mathbb{E}(Y_{bkti})$ denotes the expected adherence proportion. The intervention effect θ represents the difference in expected adherence proportion on the logit (link) scale. Because the observed adherence proportion may take values of 0 or 1, values will be rescaled to lie in the open interval (0, 1) prior to fitting the beta regression model. A beta regression precision (dispersion) parameter will be estimated as part of the model.

9.2 Correlations

We will report time-adjusted within-cluster correlations for all outcomes with 95% CIs, including the intra-cluster correlation (ICC) and within- and between-period correlations implied by the assumed correlation structures, together with all estimated variance components. For binary and ordinal outcomes we will additionally report these correlations on the latent scale; for continuous outcomes the latent and observed scales coincide.

Intra-cluster and related correlation parameters will be derived from the estimated variance components of the fitted models. For models with an AR(1) within-cluster structure, the correlation parameter will be estimated directly from the AR(1) structure. We will present these parameters in a supplementary table (Table 11).

9.3 Missing data

We will present the frequency and percentage of missing data for all variables. How we handle missing data in the primary analysis will depend on the percentage of missing primary outcome data.

If less than 10% of participants have a missing primary outcome, we will use an available-case approach. The primary unadjusted analysis will include all participants with observed primary outcome data. Adjusted analyses will be restricted to participants with complete data on the adjustment covariates, and the unadjusted analysis will be repeated in this complete-case analysis set to allow the results of the unadjusted and adjusted analyses to be directly compared.

If 10% or more of patients have a missing primary outcome, we will use multiple imputation by chained equations (MICE) under a missing-at-random assumption, imputing the primary outcome, secondary outcomes as required for analysis, and the covariates included in the fully adjusted model, with a minimum of 20 imputations. The imputation model will reflect the hierarchical data structure,

including clustering at the hospital level and design variables such as period and intervention exposure. We will then run the unadjusted and adjusted analyses on the imputed datasets, and also present the corresponding complete-case analyses for comparison. We will not impute secondary outcomes under the available-case approach.

If there is evidence that data are missing not at random, we will explore this in a sensitivity analysis.

9.4 Sensitivity analyses

We will conduct sensitivity analyses to assess how robust the primary analysis results are to the modelling assumptions, the definition of intervention exposure, and the missing-data assumption, focusing throughout on the stability of the estimated intervention effect θ . Unless otherwise stated, each sensitivity analysis uses two separate models, one with the logit link and one with the identity link. If a sensitivity model does not converge, we will report that and, where feasible, fit the next simpler random-effects or period specification, analogous to the primary-analysis convergence sequence.

9.4.1 Exposure defined by the actual date of transition

Where a cluster's actual transition date differs from its scheduled date by more than four weeks, we will refit the primary analysis model with the intervention exposure indicator X_{bkt} defined by the actual rather than the scheduled transition date. We exclude observations in the month of the actual transition, as we do for the scheduled transition, and leave all other model terms unchanged. We will report the number of clusters for which we redefine exposure.

We will explore whether models with more complex correlation structures provide a better fit to the data. These models are not used as primary analysis models because there is limited understanding of when such models converge and how best to choose between different plausible correlation structures. Instead, we include these models as sensitivity analyses.

9.4.2 Models with autoregressive within-cluster correlation

We will first explore whether allowing temporal correlation between outcomes within a cluster changes our inference about the intervention effect. Specifically, we will consider a discrete time-decay correlation structure by specifying a random cluster effect with an autoregressive structure of order one (AR(1)), described in Equation 10. This parameterisation assumes that correlations between cluster-level random effects decay geometrically with increasing temporal separation between periods.

The AR(1) process replaces the random intercept α_{bk} of the primary analysis model; all other terms, including the cluster-by-period random effect γ_{bkt} , are unchanged.

$$\alpha_{bk,t} = \rho\alpha_{bk,t-1} + \eta_{bkt}, \quad \eta_{bkt} \sim \mathcal{N}(0, \sigma_{\eta}^2) \quad (10)$$

Where:

- $\alpha_{bk,t}$: the random effect for cluster k in period t in batch b .

- ρ : the correlation between adjacent periods within the same cluster.
- η_{bkt} : a normally distributed innovation with variance σ_η^2 , which is distinct from the marginal cluster variance σ_α^2 in Equation 5.

This AR(1) formulation differs slightly from alternative parameterisations in which correlations are specified through a joint covariance matrix. Both capture decaying temporal correlation, but ρ is not directly equivalent to the parameters of those specifications and should not be interpreted interchangeably.

9.4.3 Models with random cluster by intervention effects

To assess sensitivity to the assumption of a common intervention effect across clusters, we will extend the primary analysis model to allow the intervention effect to vary by cluster. Specifically, we will include a random cluster-level slope for the intervention exposure, with a non-zero covariance between this slope and the cluster random intercept (as anticipated in the protocol). The cluster-by-period random effect remains independent of both (Equation 11).

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} + u_{bk} X_{bkt} \quad (11)$$

Where:

- u_{bk} : a random effect representing cluster-specific deviations from the average intervention effect.
- The cluster intercept and intervention slope are jointly normally distributed,

$$\begin{pmatrix} \alpha_{bk} \\ u_{bk} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_\alpha^2 & \sigma_{\alpha u} \\ \sigma_{\alpha u} & \sigma_u^2 \end{pmatrix} \right),$$

with $\sigma_{\alpha u} \neq 0$ allowed.

- $\gamma_{bkt} \sim \mathcal{N}(0, \sigma_\gamma^2)$ is independent of (α_{bk}, u_{bk}) .

9.4.4 Models with time modelled with a spline function

We will further explore the potential for a time-varying treatment effect¹⁹. To explore if the fixed period effect is both parsimonious and adequate to represent the extent of any underlying secular trend, we will model calendar time using a shared natural cubic spline basis across all batches with knots at the equally spaced time points 3, 6 and 9. This will result in five spline basis functions, because the natural cubic splines are modelled with three degrees of freedom but are constrained to be linear before the first and after the last knot. The model is specified in Equation 12.

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \sum_{j=1}^5 \beta_j S_j(t, \{3, 6, 9\}) + \theta X_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (12)$$

Where:

- t : calendar time (period index) common to all batches.
- $S_j(t, \{3, 6, 9\})$: the j th natural cubic spline basis function, shared across batches.
- β_j : the coefficient for the j th spline basis function.

9.4.5 Models exploring lag and weaning effects

We will also extend the primary analysis model with an interaction between treatment and the number of periods since first treated, to examine whether longer exposure to the intervention changes outcomes. This lets us model lag effects, where the intervention takes time to become embedded in the culture before it has an effect, and weaning effects, where its effect fades. This type of analysis attempts to disentangle how some clusters end up having a long exposure to the intervention and others have a much shorter exposure time (Equation 13).

$$g\{\Pr(Y_{bkti} = 1)\} = \mu + \beta_{bt} + \theta X_{bkt} + \theta_{\text{int}} X_{bkt} \times T_{bkt} + \alpha_{bk} + \gamma_{bkt} \quad (13)$$

Where θ_{int} represents the interaction between intervention exposure and time since first treated, and T_{bkt} is the number of periods since first exposure.

9.5 Adjusted analyses

In the fully adjusted analysis we will additionally adjust for (expected % missing data in parentheses):

- Age (<5%)
- Sex (<5%)
- Systolic blood pressure (25%)
- Glasgow Coma Scale (25%)
- Injury Severity Score (10%)
- Mechanism of injury (10%)

These are known individual-level prognostic factors for the primary outcome. These covariates will be included as fixed effects in the model selected for the primary analysis. We will model the continuous covariates assuming linear effects. We will assume no interaction between subgroups and secular time trends, which may be evaluated in future analyses if indicated.

9.6 Subgroup analyses

We will perform the following subgroup analyses of the primary outcome:

- geographical region, defined using the state in which the participating hospital is located, with each state as a separate category; the number of states depends on the final set of participating clusters and cannot be prespecified;
- age groups, defined as older adolescents (15–19 years), young adults (20–24 years), adults (25–59 years), and older adults (60 years and older)²⁰;
- sex, using the levels male and female;
- clinical cohorts, defined as blunt multisystem trauma, penetrating trauma, and severe isolated traumatic brain injury, with modification to avoid overlap between the cohorts²¹; this subgroup analysis will include only patients classified into one of these three cohorts;

- patients with major trauma, defined as patients with an injury severity score (ISS) of 16 or higher versus those with ISS below 16;
- cluster size, defined as the monthly volume of eligible patients (as used for covariate-constrained randomisation) and categorised as small (fewer than 12 patients per month), medium (12–20), or large (more than 20)

As in the protocol, we treat geographical region as our primary subgroup and the rest as exploratory.

Each subgroup analysis will use the mixed-effects binomial model selected for the primary analysis, with the logit and identity links. We will add the subgroup variable and the interaction between the subgroup variable and intervention exposure as fixed effects. From that model we will obtain the intervention effect within each subgroup level, as odds ratios and absolute risk differences, by combining the main intervention effect with the corresponding interaction term, and report each with a 95% CI. We will also report the interaction p-value as a test of heterogeneity across subgroup levels. We will assume no interaction between subgroups and secular time trends, which may be evaluated in future analyses if indicated.

9.7 Safety events

Safety events, as defined in the trial protocol, will be summarised as counts and percentages of patients with at least one recorded event, stratified by before and after ATLS training and overall. We will not adjust for clustering in these summaries. We will present these summaries in the main text. See Table 12 for details.

9.8 Statistical software

We will use the R Statistical Software for all analyses²². We will select the packages for mixed-effects modelling, small-sample corrections, multiple imputation, and reporting tables closer to the time of analysis, based on convergence behaviour and available methodology, and document them in the analysis code and final report. We will archive the analysis code used for the final analysis with the trial materials.

9.9 Presentation of analysis results

Intervention-effect estimates for the primary and secondary outcomes will be reported with 95% CIs and p-values, using the effect measures specified for each outcome. We will present key outcome analyses in the main text, and present all outcome analyses in the supplementary material. See Table 8 for key analyses and Table 9 for all analyses.

Sensitivity, fully adjusted, and subgroup analyses of the primary outcome will be reported in the same style, in the supplementary material. See Table 10. Subgroup-specific intervention effects will also be presented in the main text as a forest plot with 95% CIs for the odds ratio and absolute risk difference. See Figure 7.

10 Shell figures and tables

The captions in this section state whether each table or figure is intended for the main text or the supplementary material of the trial report.

10.1 CONSORT diagrams

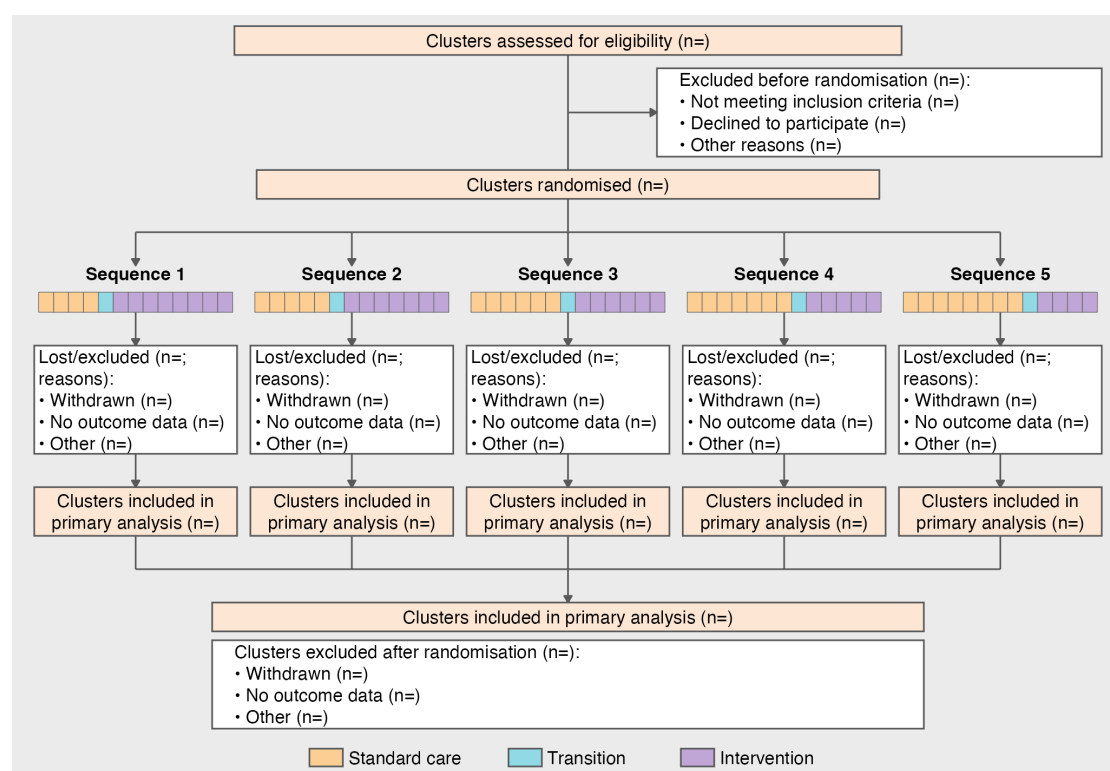


Figure 4: Cluster-level CONSORT diagram shell. Placeholders cover clusters assessed and randomised, losses and exclusions after randomisation with reasons, and clusters included in the primary analysis, by sequence and overall. The flowchart summarises data across all batches 1-6. Intended for the main text.

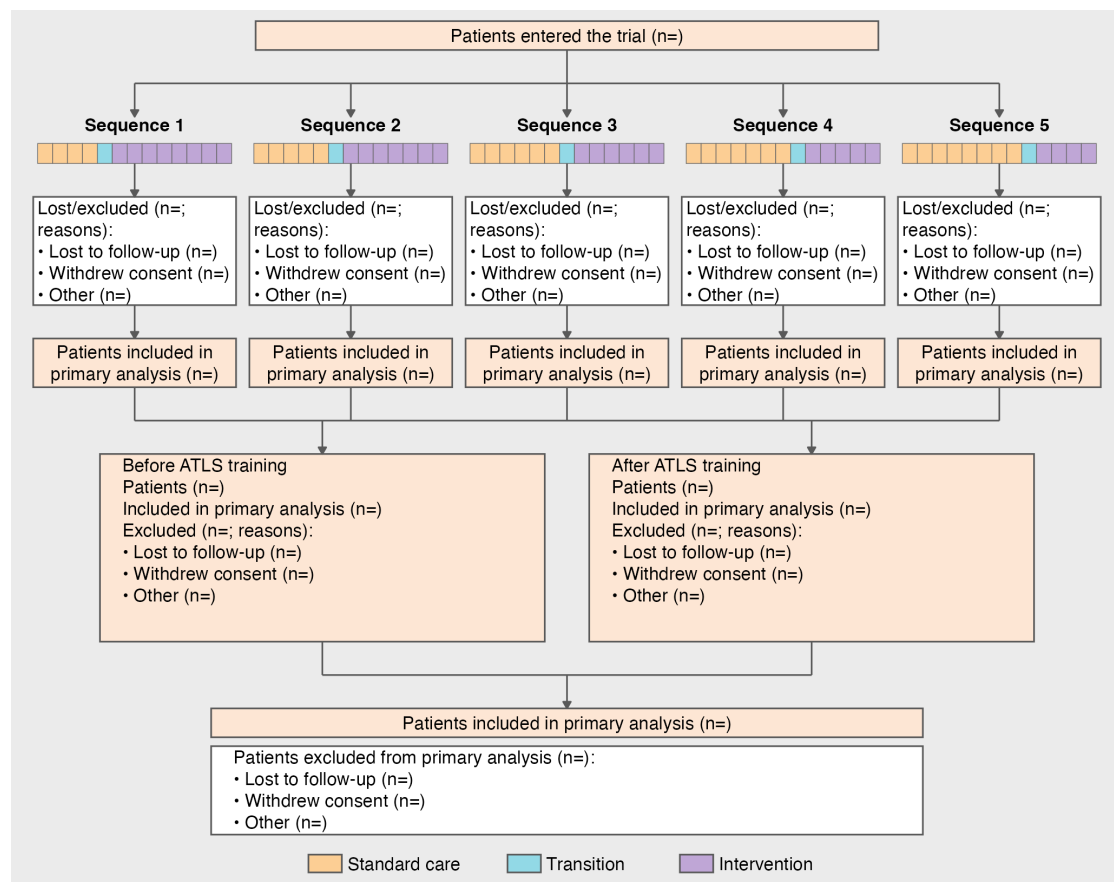


Figure 5: Patient-level CONSORT diagram shell. Placeholders cover patients entered, losses and exclusions from the primary analysis with reasons, and patients analysed, by sequence, before and after ATLS training, and overall. The flowchart summarises data across all batches 1-6. Intended for the main text.

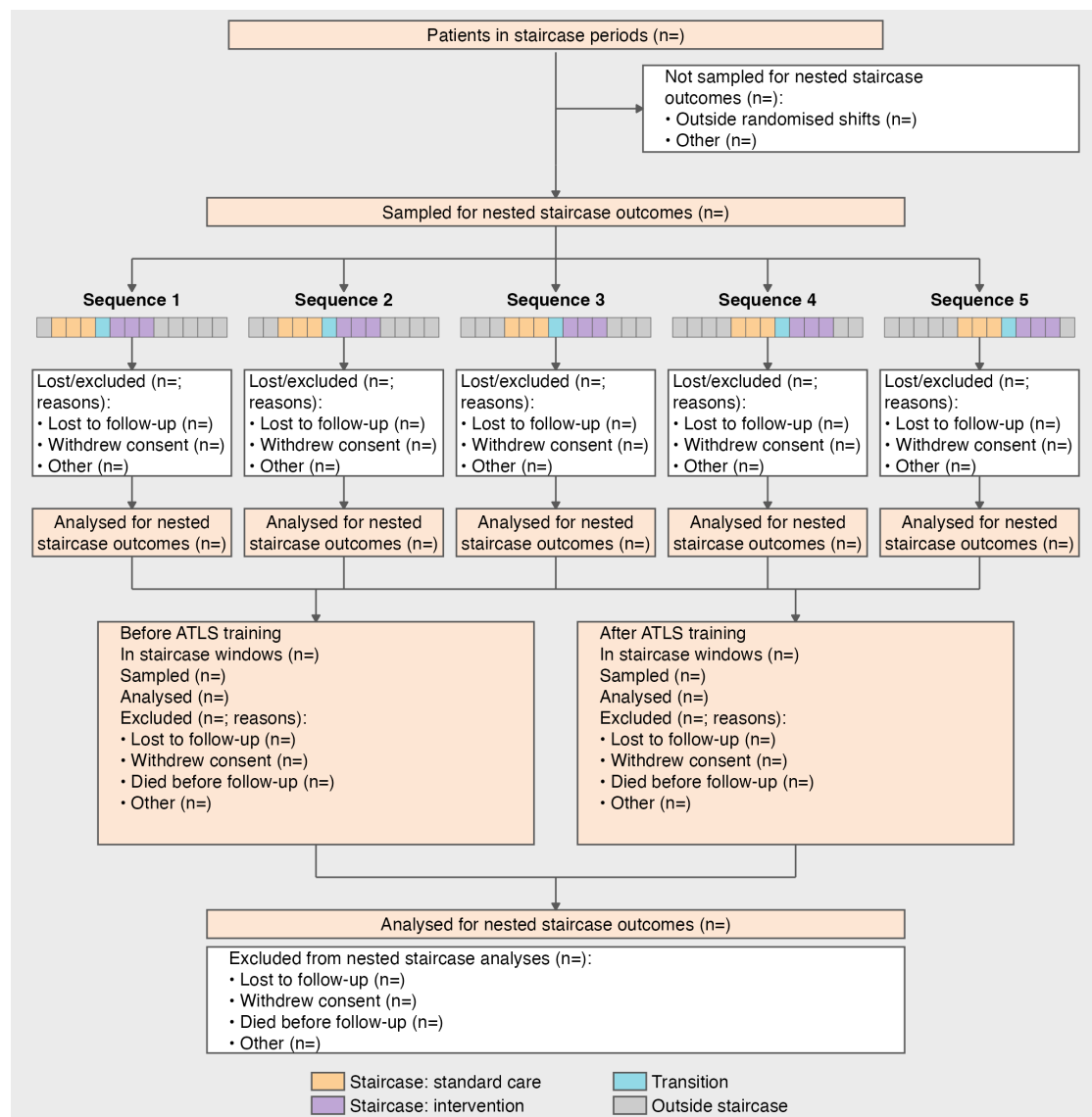


Figure 6: Nested-staircase CONSORT diagram shell. Placeholders cover patients in staircase periods, the shift-stratified nested sample, losses and exclusions with reasons, and patients analysed for nested-staircase secondary outcomes, by sequence, before and after ATLS training, and overall. Coloured strip cells mark staircase months; grey cells are outside nested data collection. The flowchart summarises data across all batches 1-6. Intended for the supplementary material.

10.2 Shell tables and figures for analysis results

Table 2: Cluster characteristics. Intended for the main text.

Characteristic	Implementation sequence					Overall
	Sequence 1	Sequence 2	Sequence 3	Sequence 4	Sequence 5	
State, n (%)						
States depend on participating clusters						
Monthly trauma patient volume, Median (Q1, Q3)						
Hospital beds, Median (Q1, Q3)						
Intensive care unit beds, Median (Q1, Q3)						
Dedicated trauma beds, Median (Q1, Q3)						
Specialities available around the clock, n (%)						
General surgery						
Orthopaedics						
Neurosurgery						
Vascular surgery						
Interventional radiology						
Emergency medicine						
Facilities available around the clock, n (%)						
X-ray						
Ultrasound/FAST						
CT						
MRI						
Blood bank						
Initial resuscitation provider, n (%)						
Casualty medical officers						
Emergency medicine residents						
Surgical residents						

Table 3: Key patient characteristics. Intended for the main text.

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Age (years), Median (Q1, Q3)			
Missing			
Sex, n (%)			
Female			
Male			
Other			
Missing			
Transferred in, n (%)			
Missing			
Mechanism of injury, n (%)			
Road traffic injury			
Fall			
Assault			
Other			
Missing			

(continued)

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Injury Severity Score, Median (Q1, Q3)			
Missing			
Glasgow Coma Scale score, Median (Q1, Q3)			
Missing			
Systolic blood pressure (mmHg), Median (Q1, Q3)			
Missing			
Respiratory rate (breaths/min), Median (Q1, Q3)			
Missing			
Oxygen saturation (%), Median (Q1, Q3)			
Missing			
Surgery, n (%)			
Missing			
Transfusion, n (%)			
Missing			
Imaging, n (%)			
Ultrasound			
X-ray			
CT			
Missing			

Table 4: All patient characteristics. Intended for the supplementary material.

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Age (years), Median (Q1, Q3)			
Missing			
Sex, n (%)			
Female			
Male			
Other			
Missing			
Marital status, n (%)			
Never married			
Currently married			
Separated			
Divorced			
Widowed			
Cohabiting			
Missing			
Education level, n (%)			
Not attended school			
Primary school			
Secondary school			
Higher secondary school			
Graduate			
Post graduate and above			
Other			
Missing			
Main work status, n (%)			

(continued)

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Paid work, such as daily wage earner, teacher, factory worker and government employee			
Self-employed, such as own your business or farming			
Non-paid work, such as volunteer or charity			
Student			
Keeping house/homemaker			
Retired			
Unemployed (health reasons)			
Unemployed (other reasons)			
Other			
No income			
Missing			
Income level (INR per month), n (%)			
Below 10,000			
10,001-20,000			
20,001-30,000			
30,001-50,000			
50,001-80,000			
80,001-1,00,000			
Above 1,00,000			
No income			
Missing			
Comorbidities (Charlson Comorbidity Index), n (%)			
Myocardial infarction			
Congestive heart failure			
Peripheral vascular disease			
Cerebrovascular disease			
Dementia			
Chronic pulmonary disease			
Rheumatologic disease			
Peptic ulcer disease			
Liver disease			
Diabetes			
Hemiplegia or paraplegia			
Renal disease			
Malignancy			
Leukemia			
Lymphoma			
AIDS			
None			
Other			
Missing			
Severity of liver disease, n (%)			
None			
Mild			
Moderate or severe			
Missing			
Severity of diabetes, n (%)			
None			
Controlled			
Uncontrolled			
Missing			

(continued)

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Severity of malignancy, n (%)			
None			
Localized			
Metastatic tumor			
Missing			
Clinical Frailty Scale, n (%)			
1. Very Fit			
2. Fit			
3. Managing well			
4. Living with very mild frailty			
5. Living with mild frailty			
6. Living with moderate frailty			
7. Living with severe frailty			
8. Living with very severe frailty			
9. Terminally ill			
Missing			
Mode of transport, n (%)			
Ambulance			
Police			
Private vehicle			
Walking			
Others			
Missing			
Transferred in, n (%)			
Missing			
Mechanism of injury, n (%)			
Road traffic injury			
Fall			
Assault			
Other			
Missing			
Injury Severity Score, Median (Q1, Q3)			
Missing			
Injury source data, n (%)			
Medical record			
X-ray report			
CT-report			
Surgical notes			
Missing			
Glasgow Coma Scale score, Median (Q1, Q3)			
Missing			
Systolic blood pressure (mmHg), Median (Q1, Q3)			
Missing			
Diastolic blood pressure (mmHg), Median (Q1, Q3)			
Missing			
Heart rate (beats/min), Median (Q1, Q3)			
Missing			
Respiratory rate (breaths/min), Median (Q1, Q3)			
Missing			
Oxygen saturation (%), Median (Q1, Q3)			
Missing			

(continued)

Characteristic	ATLS training		Overall
	Before ATLS	After ATLS	
Body temperature (°F), Median (Q1, Q3)			
Missing			
Emergency department disposition, n (%)			
Admitted			
Referred or transferred for admission			
Dead			
Others			
Missing			
Type of admitting ward, n (%)			
General surgery			
Orthopaedics			
Neurosurgery			
Intensive care unit			
High dependency unit			
Medicine			
Trauma ward			
Missing			
Hospital disposition, n (%)			
Alive			
Dead			
Transferred for admission			
Missing			
Transferred to another hospital, n (%)			
Missing			
Surgery, n (%)			
Missing			
Preoperative ASA score, n (%)			
1. A normal healthy patient			
2. A patient with mild systemic disease			
3. A patient with severe systemic disease			
4. A patient with severe systemic disease that is a constant threat to life			
5. A moribund patient who is not expected to survive without the operation			
6. A declared brain-dead patient whose organs are being removed for donor purposes			
Missing			
Transfusion, n (%)			
Missing			
Type of blood product, n (%)			
Packed red blood cells			
Platelets			
Fresh frozen plasma			
Whole blood			
Other			
Missing			
Number of units transfused, Median (Q1, Q3)			
Missing			
Imaging, n (%)			
Ultrasound			
X-ray			
CT			
Missing			

Table 5: Intercurrent events by outcome. Intended for the supplementary material.

Intercurrent event	ATLS training		Overall
	Before ATLS	After ATLS	
Primary outcome			
In-hospital mortality within 30 days			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
All-cause mortality within 30 days			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
All-cause mortality within three months			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Length of emergency department stay			
Death, n (%)			
Transfer to another hospital, n (%)			
Length of hospital stay			
Death, n (%)			
Transfer to another hospital, n (%)			
Intensive care unit (ICU) admission			
Death, n (%)			
Transfer to another hospital, n (%)			
Length of intensive care unit stay			
Death, n (%)			
Transfer to another hospital, n (%)			
Return to work at 30 days			
Death before 30 days, n (%)			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Return to work at three months			
Death before three months, n (%)			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS® principles			
Incomplete observation of resuscitation, n (%)			
Secondary outcomes within seven days of discharge (nested staircase design)			
Quality of life			
Death before assessment, n (%)			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Disability			
Death before assessment, n (%)			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
Quality of life			
Death before 30 days, n (%)			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Disability			
Death before 30 days, n (%)			

(continued)

Intercurrent event	ATLS training		Overall
	Before ATLS	After ATLS	
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
Quality of life			
Death before three months, n (%)			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			
Disability			
Death before three months, n (%)			
Withdrawal of consent for non-routine data collection, n (%)			
Loss to follow-up, n (%)			

Table 6: Descriptive summaries of key outcomes. Intended for the main text.

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
Primary outcome			
In-hospital mortality within 30 days, n (%)			
Missing			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours, n (%)			
Missing			
All-cause mortality within 30 days, n (%)			
Missing			
All-cause mortality within three months, n (%)			
Missing			
Length of emergency department stay (hours), Median (Q1, Q3)			
Missing			
Length of hospital stay (days), Median (Q1, Q3)			
Missing			
Intensive care unit admission, n (%)			
Missing			
Length of intensive care unit stay (days), Median (Q1, Q3)			
Missing			
Return to work at 30 days, n (%)			
Missing			
Return to work at three months, n (%)			
Missing			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS principles (%), Median (Q1, Q3)			
Missing			
Secondary outcomes within seven days of discharge (nested staircase design)			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			

Table 7: Descriptive summaries of all outcomes. Intended for the supplementary material.

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
Primary outcome			
In-hospital mortality within 30 days, n (%)			
Missing			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours, n (%)			
Missing			
All-cause mortality within 30 days, n (%)			
Missing			
All-cause mortality within three months, n (%)			
Missing			
Length of emergency department stay (hours), Median (Q1, Q3)			
Missing			
Length of hospital stay (days), Median (Q1, Q3)			
Missing			
Intensive care unit admission, n (%)			
Missing			
Length of intensive care unit stay (days), Median (Q1, Q3)			
Missing			
Return to work at 30 days, n (%)			
Missing			
Return to work at three months, n (%)			
Missing			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS principles (%), Median (Q1, Q3)			
Missing			
Secondary outcomes within seven days of discharge (nested staircase design)			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
Missing			
EQ-5D-5L mobility, n (%)			
1. I have no problems in walking about			
2. I have slight problems in walking about			
3. I have moderate problems in walking about			
4. I have severe problems in walking about			
5. I am unable to walk about			
Missing			
EQ-5D-5L self-care, n (%)			
1. I have no problems washing or dressing myself			
2. I have slight problems washing or dressing myself			
3. I have moderate problems washing or dressing myself			
4. I have severe problems washing or dressing myself			
5. I am unable to wash or dress myself			
Missing			
EQ-5D-5L usual activities, n (%)			
1. I have no problems doing my usual activities			
2. I have slight problems doing my usual activities			
3. I have moderate problems doing my usual activities			
4. I have severe problems doing my usual activities			
5. I am unable to do my usual activities			
Missing			
EQ-5D-5L pain/discomfort, n (%)			
1. I have no pain or discomfort			
2. I have slight pain or discomfort			
3. I have moderate pain or discomfort			
4. I have severe pain or discomfort			
5. I have extreme pain or discomfort			
Missing			
EQ-5D-5L anxiety/depression, n (%)			
1. I am not anxious or depressed			
2. I am slightly anxious or depressed			
3. I am moderately anxious or depressed			
4. I am severely anxious or depressed			
5. I am extremely anxious or depressed			
Missing			
WHODAS 2.0 cognition, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 mobility, n (%)			
1. None			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 self-care, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 getting along, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 life activities, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 participation, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			
EQ-5D-5L mobility, n (%)			
1. I have no problems in walking about			
2. I have slight problems in walking about			
3. I have moderate problems in walking about			
4. I have severe problems in walking about			
5. I am unable to walk about			
Missing			
EQ-5D-5L self-care, n (%)			
1. I have no problems washing or dressing myself			
2. I have slight problems washing or dressing myself			
3. I have moderate problems washing or dressing myself			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
4. I have severe problems washing or dressing myself			
5. I am unable to wash or dress myself			
Missing			
EQ-5D-5L usual activities, n (%)			
1. I have no problems doing my usual activities			
2. I have slight problems doing my usual activities			
3. I have moderate problems doing my usual activities			
4. I have severe problems doing my usual activities			
5. I am unable to do my usual activities			
Missing			
EQ-5D-5L pain/discomfort, n (%)			
1. I have no pain or discomfort			
2. I have slight pain or discomfort			
3. I have moderate pain or discomfort			
4. I have severe pain or discomfort			
5. I have extreme pain or discomfort			
Missing			
EQ-5D-5L anxiety/depression, n (%)			
1. I am not anxious or depressed			
2. I am slightly anxious or depressed			
3. I am moderately anxious or depressed			
4. I am severely anxious or depressed			
5. I am extremely anxious or depressed			
Missing			
WHODAS 2.0 cognition, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 mobility, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 self-care, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 getting along, n (%)			
1. None			
2. Mild			
3. Moderate			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 life activities, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 participation, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
EQ-5D-5L VAS, Median (Q1, Q3)			
Missing			
WHODAS 2.0 summary score, Median (Q1, Q3)			
Missing			
EQ-5D-5L mobility, n (%)			
1. I have no problems in walking about			
2. I have slight problems in walking about			
3. I have moderate problems in walking about			
4. I have severe problems in walking about			
5. I am unable to walk about			
Missing			
EQ-5D-5L self-care, n (%)			
1. I have no problems washing or dressing myself			
2. I have slight problems washing or dressing myself			
3. I have moderate problems washing or dressing myself			
4. I have severe problems washing or dressing myself			
5. I am unable to wash or dress myself			
Missing			
EQ-5D-5L usual activities, n (%)			
1. I have no problems doing my usual activities			
2. I have slight problems doing my usual activities			
3. I have moderate problems doing my usual activities			
4. I have severe problems doing my usual activities			
5. I am unable to do my usual activities			
Missing			
EQ-5D-5L pain/discomfort, n (%)			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
1. I have no pain or discomfort			
2. I have slight pain or discomfort			
3. I have moderate pain or discomfort			
4. I have severe pain or discomfort			
5. I have extreme pain or discomfort			
Missing			
EQ-5D-5L anxiety/depression, n (%)			
1. I am not anxious or depressed			
2. I am slightly anxious or depressed			
3. I am moderately anxious or depressed			
4. I am severely anxious or depressed			
5. I am extremely anxious or depressed			
Missing			
WHODAS 2.0 cognition, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 mobility, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 self-care, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 getting along, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 life activities, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			
Missing			
WHODAS 2.0 participation, n (%)			
1. None			
2. Mild			
3. Moderate			
4. Severe			
5. Extreme or cannot do			

(continued)

Outcome	ATLS training		Overall
	Before ATLS	After ATLS	
Missing			

Table 8: Shell table of key primary and secondary outcome analysis results. Intended for the main text.

Outcome	Estimate	95% CI	p-value
Primary outcome			
In-hospital mortality within 30 days			
Odds ratio			
Absolute risk difference			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours			
Odds ratio			
Absolute risk difference			
All-cause mortality within 30 days			
Odds ratio			
Absolute risk difference			
All-cause mortality within three months			
Odds ratio			
Absolute risk difference			
Length of emergency department stay			
Rate ratio			
Length of hospital stay			
Rate ratio			
Intensive care unit (ICU) admission			
Odds ratio			
Absolute risk difference			
Length of intensive care unit stay			
Rate ratio			
Return to work at 30 days			
Odds ratio			
Absolute risk difference			
Return to work at three months			
Odds ratio			
Absolute risk difference			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS® principles			
Logit-scale difference			
Secondary outcomes within seven days of discharge (nested staircase design)			
Quality of life			
Cumulative odds ratio			
Mean difference			
Disability			
Cumulative odds ratio			
Mean difference			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
Quality of life			
Cumulative odds ratio			
Mean difference			
Disability			
Cumulative odds ratio			
Mean difference			

(continued)

Outcome	Estimate	95% CI	p-value
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
Quality of life			
Cumulative odds ratio			
Mean difference			
Disability			
Cumulative odds ratio			
Mean difference			
Abbreviations: CI = Confidence Interval, IRR = Incidence Rate Ratio, OR = Odds Ratio			

Table 9: Shell table of all primary and secondary outcome analysis results. Intended for the supplementary material.

Outcome	Estimate	95% CI	p-value
Primary outcome			
In-hospital mortality within 30 days			
Odds ratio			
Absolute risk difference			
Secondary outcomes (main stepped-wedge design)			
All-cause mortality within 24 hours			
Odds ratio			
Absolute risk difference			
All-cause mortality within 30 days			
Odds ratio			
Absolute risk difference			
All-cause mortality within three months			
Odds ratio			
Absolute risk difference			
Length of emergency department stay			
Rate ratio			
Length of hospital stay			
Rate ratio			
Intensive care unit (ICU) admission			
Odds ratio			
Absolute risk difference			
Length of intensive care unit stay			
Rate ratio			
Return to work at 30 days			
Odds ratio			
Absolute risk difference			
Return to work at three months			
Odds ratio			
Absolute risk difference			
Secondary outcomes during initial resuscitation (nested staircase design)			
Adherence to ATLS® principles			
Logit-scale difference			
Secondary outcomes within seven days of discharge (nested staircase design)			
EQ-5D-5L mobility			
Cumulative odds ratio			
EQ-5D-5L self-care			
Cumulative odds ratio			
EQ-5D-5L usual activities			
Cumulative odds ratio			
EQ-5D-5L pain/discomfort			

(continued)

Outcome	Estimate	95% CI	p-value
Cumulative odds ratio			
EQ-5D-5L anxiety/depression			
Cumulative odds ratio			
EQ-5D-5L VAS			
Mean difference			
WHODAS 2.0 cognition			
Cumulative odds ratio			
WHODAS 2.0 mobility			
Cumulative odds ratio			
WHODAS 2.0 self-care			
Cumulative odds ratio			
WHODAS 2.0 getting along			
Cumulative odds ratio			
WHODAS 2.0 life activities			
Cumulative odds ratio			
WHODAS 2.0 participation			
Cumulative odds ratio			
WHODAS 2.0 summary score			
Mean difference			
Secondary outcomes at 30 days after arrival at the emergency department (nested staircase design)			
EQ-5D-5L mobility			
Cumulative odds ratio			
EQ-5D-5L self-care			
Cumulative odds ratio			
EQ-5D-5L usual activities			
Cumulative odds ratio			
EQ-5D-5L pain/discomfort			
Cumulative odds ratio			
EQ-5D-5L anxiety/depression			
Cumulative odds ratio			
EQ-5D-5L VAS			
Mean difference			
WHODAS 2.0 cognition			
Cumulative odds ratio			
WHODAS 2.0 mobility			
Cumulative odds ratio			
WHODAS 2.0 self-care			
Cumulative odds ratio			
WHODAS 2.0 getting along			
Cumulative odds ratio			
WHODAS 2.0 life activities			
Cumulative odds ratio			
WHODAS 2.0 participation			
Cumulative odds ratio			
WHODAS 2.0 summary score			
Mean difference			
Secondary outcomes at three months after arrival at the emergency department (nested staircase design)			
EQ-5D-5L mobility			
Cumulative odds ratio			
EQ-5D-5L self-care			
Cumulative odds ratio			
EQ-5D-5L usual activities			
Cumulative odds ratio			
EQ-5D-5L pain/discomfort			
Cumulative odds ratio			
EQ-5D-5L anxiety/depression			

(continued)

Outcome	Estimate	95% CI	p-value
Cumulative odds ratio			
EQ-5D-5L VAS			
Mean difference			
WHODAS 2.0 cognition			
Cumulative odds ratio			
WHODAS 2.0 mobility			
Cumulative odds ratio			
WHODAS 2.0 self-care			
Cumulative odds ratio			
WHODAS 2.0 getting along			
Cumulative odds ratio			
WHODAS 2.0 life activities			
Cumulative odds ratio			
WHODAS 2.0 participation			
Cumulative odds ratio			
WHODAS 2.0 summary score			
Mean difference			
Abbreviations: CI = Confidence Interval, IRR = Incidence Rate Ratio, OR = Odds Ratio			

Table 10: Shell table of sensitivity, fully adjusted, and subgroup analysis results for the primary outcome. Intended for the supplementary material.

Analysis	Estimate	95% CI	p-value
Sensitivity analyses			
Autoregressive within-cluster correlation (AR(1))			
Odds ratio			
Absolute risk difference			
Random cluster-by-intervention effects			
Odds ratio			
Absolute risk difference			
Time modelled with a spline function			
Odds ratio			
Absolute risk difference			
Lag and weaning effects (intervention)			
Odds ratio			
Absolute risk difference			
Lag and weaning effects (periods since first exposure)			
Odds ratio			
Absolute risk difference			
Actual date of transition for intervention exposure			
Odds ratio			
Absolute risk difference			
Fully adjusted analysis			
Odds ratio			
Absolute risk difference			
Subgroup analyses			
Geographical region (state-specific estimates; states depend on participating clusters)			
Odds ratio			
Absolute risk difference			
Age group: Older adolescents (15-19 years)			
Odds ratio			
Absolute risk difference			

(continued)

Analysis	Estimate	95% CI	p-value
Age group: Young adults (20-24 years)			
Odds ratio			
Absolute risk difference			
Age group: Adults (25-59 years)			
Odds ratio			
Absolute risk difference			
Age group: Older adults (60 years and older)			
Odds ratio			
Absolute risk difference			
Sex: Male			
Odds ratio			
Absolute risk difference			
Sex: Female			
Odds ratio			
Absolute risk difference			
Clinical cohort: Blunt multisystem trauma			
Odds ratio			
Absolute risk difference			
Clinical cohort: Penetrating trauma			
Odds ratio			
Absolute risk difference			
Clinical cohort: Severe isolated traumatic brain injury			
Odds ratio			
Absolute risk difference			
Major trauma: ISS ≥ 16			
Odds ratio			
Absolute risk difference			
Major trauma: ISS < 16			
Odds ratio			
Absolute risk difference			
Cluster size: Small (< 12 patients/month)			
Odds ratio			
Absolute risk difference			
Cluster size: Medium (12-20 patients/month)			
Odds ratio			
Absolute risk difference			
Cluster size: Large (> 20 patients/month)			
Odds ratio			
Absolute risk difference			
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Table 11: Shell table of within-cluster correlation parameters for all outcomes. Intended for the supplementary material.

Parameter	Estimate	95% CI
In-hospital mortality within 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		

(continued)

Parameter	Estimate	95% CI
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
All-cause mortality within 24 hours		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
All-cause mortality within 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
All-cause mortality within three months		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Length of emergency department stay		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Length of hospital stay		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Intensive care unit (ICU) admission		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		

(continued)

Parameter	Estimate	95% CI
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Length of intensive care unit stay		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Return to work at 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Return to work at three months		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Adherence to ATLS® principles		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Quality of life within seven days of discharge		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		

(continued)

Parameter	Estimate	95% CI
Quality of life at 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Quality of life at three months		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Disability within seven days of discharge		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Disability at 30 days		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Disability at three months		
Cluster variance		
Cluster-by-period variance		
Intra-cluster correlation (ICC)		
Within-period correlation		
Between-period correlation		
Latent-scale ICC		
Latent-scale within-period correlation		
Latent-scale between-period correlation		
AR(1) within-period correlation		
AR(1) correlation decay parameter (ρ)		
Abbreviation: CI = Confidence Interval		

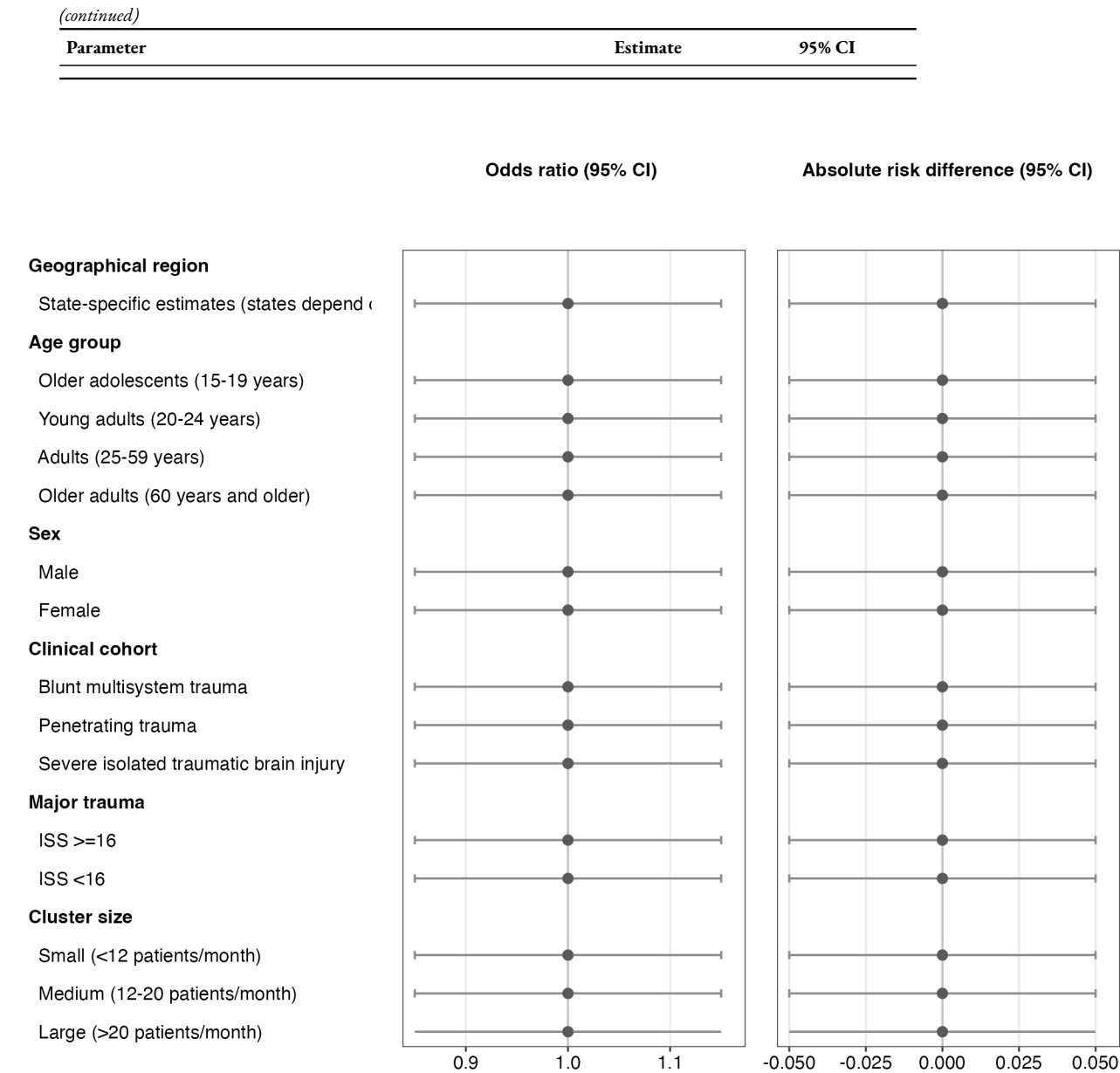


Table 12: Safety events. Intended for the main text.

Safety event	ATLS training		Overall
	Before ATLS	After ATLS	
Prolonged mechanical ventilation (>7 days), n (%)			
Initiation of renal replacement therapy, n (%)			
Prolonged (>2 days) or renewed use of vasopressors such as norepinephrine or vasopressin, n (%)			
Other reported safety events, n (%)			
Events assessed as probably related to the trial or intervention, n (%)			

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